

Sathishkumar Ranganathan

FULL-STACK SOFTWARE ENGINEER | AI APPLICATION DEVELOPMENT | BACKEND & CLOUD

Singapore | Open to India & International Opportunities

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Professional Summary

Full-Stack Software Engineer with 10+ years of IT experience and 5+ years of hands-on experience designing, developing, and deploying scalable web, mobile, and cloud applications. Strong expertise in Laravel, React.js, Node.js, Flutter, RESTful APIs, AWS, database optimization, secure application architecture, and production deployment. Currently expanding into Generative AI and AI application development, with a focus on LLM-based applications, AI API integration, RAG, embeddings, and AI-powered enterprise solutions. Experienced across the full software development lifecycle, from requirements and architecture through development, deployment, maintenance, and production support.

Core Technical Skills

AI & Generative AI	: Generative AI, LLM Application Development, Prompt Engineering, AI API Integration, RAG, Embeddings, Vector Databases
Languages	: Python, PHP, JavaScript, C#, VB.NET
Frontend	: React.js, Next.js, HTML5, CSS3, Bootstrap
Backend & APIs	: Laravel, Node.js, Express.js, CodeIgniter, RESTful API Development
Mobile Development	: Flutter, React Native
Databases	: MySQL, Microsoft SQL Server, MongoDB, SQLite, Database Optimization & Data Modelling
Cloud & DevOps	: AWS (EC2, RDS, S3, IAM), Vercel, CI/CD Pipelines, Cloud Deployment
Architecture & Security	: Scalable Backend Architecture, Full-Stack Application Design, Performance Optimization, JWT, RBAC, API Security, Secure System Development
Tools	: Git, GitHub, Bitbucket, Postman, VS Code, Android Studio, NPM, Composer, Figma, Codemagic
Practices	: Agile & Scrum, Version Control, System Maintenance, Troubleshooting, SDLC

Professional Experience

Software Developer | Pedro, Singapore | Sep 2020 – Present

- Designed and delivered enterprise workforce, security, access-control, and visitor-management solutions using Laravel, React.js, Flutter, MySQL, and AWS.
- Architected scalable RESTful APIs with secure authentication, authorization, JWT, and role-based access control (RBAC) for integrated web and mobile platforms.
- Built and deployed cloud-native applications on AWS with secure infrastructure configurations and production deployment workflows.
- Developed offline-first mobile modules with background synchronization for operations in low-network environments.
- Optimized backend services, database queries, indexing, and application workflows to improve performance, API responsiveness, and reliability.
- Managed the full SDLC including requirements analysis, system architecture, development, testing, deployment, maintenance, and production support.

- Integrated third-party services including payment gateways, email, QR authentication, push notifications, and cloud storage.

IT Support Engineer | Digital Alarm Technologies, Singapore | Mar 2018 – Jul 2020

- Supported enterprise Windows/Linux infrastructure and applications, maintaining system stability, security, and operational continuity.
- Participated in Windows Server migration projects and managed patching, upgrades, backups, monitoring, and preventive maintenance.
- Provided Tier 2 support across application, server, hardware, and network issues and collaborated with vendors and internal teams on technical incidents.

Support Engineer | Hinduja Global Solutions, India | Jun 2016 – Jan 2018

- Provided remote and on-site technical support for hardware, software, operating systems, printers, and network connectivity.
- Managed workstation setup, software configuration, user onboarding, troubleshooting, and technical documentation.

Support Engineer | KBG Services, India | Jun 2015 – May 2016

- Maintained Windows server, desktop, and network services and supported hardware diagnostics, software installation, backups, and connectivity troubleshooting.

Selected Projects

iSecure360 – Workforce & Security Management Platform

Client Location: Singapore | **Technologies:** Laravel, MySQL, Flutter, AWS

Summary: Enterprise workforce and security operations platform designed for security agencies and facility management operations

- Architected and developed an enterprise-grade workforce and security management platform with web and cross-platform mobile applications.
- Implemented modules like attendance management, incident reporting, leave management, patrol checkpoints, and operational checklists and GPS-based guard tour tracking with offline-first mobile capabilities and background synchronization.
- Designed secure RESTful APIs with encrypted communication, JWT authentication, and role-based access control (RBAC).
- Optimized backend performance, database queries, and application workflows for high availability and scalability.
- Deployed and maintained AWS cloud infrastructure including EC2, RDS, S3, and automated backup configurations.

iSecure360 – Visitor Management System

Client Location: Singapore | **Technologies:** Laravel, MySQL, Flutter, AWS

Summary: Digital visitor and contractor management platform built for secure access control and streamlined visitor operations.

- Designed secure digital visitor check-in/check-out workflows and QR-based authentication and identity-verification processes.
- Built real-time visitor monitoring dashboards with live status tracking and reporting.
- Developed scalable REST APIs for web and mobile integration and optimized database performance for high-concurrency workloads.
- Integrated notification, email, and cloud-storage services for operational automation.

AVMS – Access Control & Visitor Management System

Client Location: Singapore | **Technologies:** React.js, MySQL, Flutter, AWS

Summary: Smart residential access control and visitor management solution integrated with automated entry and monitoring systems.

- Architected React.js web and Flutter mobile applications for visitor pre-registration, facility booking, resident feedback, and access approvals.
- Integrated ANPR and smart door-access systems for automated residential community entry management.
- Built secure REST APIs with encrypted data transmission, authentication, and role-based authorization.
- Managed AWS EC2, RDS, S3, and IAM with security-focused deployment configurations.
- Improved performance through database indexing, query optimization, and backend caching.

Additional projects available upon request or via GitHub/Portfolio

AI Learning & Current Development

- Currently developing skills in Generative AI and AI application development, including LLM concepts, prompt engineering, AI API integration, RAG, embeddings, and vector-based retrieval.
- Applying existing backend, API, security, database, and cloud engineering experience toward production-oriented AI application architecture.
- Building practical AI projects to demonstrate integration of AI capabilities into enterprise web applications.

Education

Bachelor of Engineering (Computer Science and Engineering)

Panimalar Engineering College, Anna University - Chennai, Tamil Nadu, India - (2011 – 2015)

Certifications

- ITIL Foundation – PeopleCert
- Google IT Support Professional Certificate – Coursera
- Full Stack Web Development – Udemy
- Mobile App Development (Flutter) – Udemy
- Python Programming - Skillup